WORKED SOLUTION

STEP I Specimen paper, Question 3

Pure • Second Order Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q3

The key idea

A second-order equation is hard; two first-order equations are easy. The suggested substitution $z = y' - y$ (writing y' for $\mathrm{d}y/\mathrm{d}x$) is chosen so that the left-hand side collapses into an expression involving only z and z'. That leaves a first-order linear equation for z, and then a second first-order linear equation for y; each yields to an integrating factor.

Solution

Reducing the order

With $z = y' - y$ we have $z' = y'' - y'$. We split the left-hand side of the equation to match:

$$y'' + 2y' - 3y = (y'' - y') + 3(y' - y) = z' + 3z.$$

So the equation becomes

$$\frac{\mathrm{d}z}{\mathrm{d}x} + 3z = 2\mathrm{e}^x.$$

The initial conditions give $z(0) = y'(0) - y(0) = 2 - 1 = 1$.

? **Ask yourself** How would you have spotted the split $(y'' - y') + 3(y' - y)$?

We need $y'' + 2y' - 3y$ to equal $z' + kz = y'' + (k-1)y' - ky$ for some constant k. Matching the y' terms gives $k = 3$, and then $-ky = -3y$ matches too. (This works because 1 is a root of the auxiliary equation $m^2 + 2m - 3 = (m-1)(m+3) = 0$; the other root, -3, is what appears in $z' + 3z$.)

Solving for z

Multiplying by the integrating factor e^{3x},

$$\frac{\mathrm{d}}{\mathrm{d}x}(z\mathrm{e}^{3x}) = 2\mathrm{e}^{4x} \quad \Longrightarrow \quad z\mathrm{e}^{3x} = \tfrac{1}{2}\mathrm{e}^{4x} + A.$$

At $x = 0$, $1 = \frac{1}{2} + A$, so $A = \frac{1}{2}$ and

$$z = \tfrac{1}{2}(\mathrm{e}^{x} + \mathrm{e}^{-3x}).$$

Solving for y

Now $y' - y = z$, that is,

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y = \tfrac{1}{2}(\mathrm{e}^{x} + \mathrm{e}^{-3x}).$$

The integrating factor is e^{-x}:

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^{-x}) = \tfrac{1}{2} + \tfrac{1}{2}\mathrm{e}^{-4x},$$
$$y\mathrm{e}^{-x} = \tfrac{1}{2}x - \tfrac{1}{8}\mathrm{e}^{-4x} + B.$$

At $x = 0$, $1 = -\frac{1}{8} + B$, so $B = \frac{9}{8}$. Multiplying through by e^{x},

$$\boxed{y = \left(\frac{x}{2} + \frac{9}{8}\right)\mathrm{e}^{x} - \frac{1}{8}\mathrm{e}^{-3x} = \frac{(4x+9)\mathrm{e}^{x} - \mathrm{e}^{-3x}}{8}}$$

Notice that we used only $y(0) = 1$ here; the condition $y'(0) = 2$ was already used in finding $z(0)$, so both conditions have been applied.

Checks

- $y(0) = (9-1)/8 = 1$, and $y' = \left((4x+13)\mathrm{e}^{x} + 3\mathrm{e}^{-3x}\right)/8$ gives $y'(0) = 16/8 = 2$.
- Substituting: e^{-3x} solves the homogeneous equation ($9 - 6 - 3 = 0$), and so does e^{x} ($1 + 2 - 3 = 0$). For $x\mathrm{e}^{x}$, $y'' + 2y' - 3y = (x+2) + 2(x+1) - 3x = 4$ times e^{x}, so $\frac{1}{2}x\mathrm{e}^{x}$ contributes exactly $2\mathrm{e}^{x}$.
- Numerically, stepping the original equation from $x = 0$ with a Runge–Kutta method gives $y(1) \approx 4.41098$, agreeing with $(13\mathrm{e} - \mathrm{e}^{-3})/8$.

Common slips

- Applying $y'(0) = 2$ a second time at the end, or forgetting to convert it into $z(0) = 1$.
- Trying a particular integral $c\,\mathrm{e}^{x}$ "otherwise": because e^{x} already solves the homogeneous equation this fails, and one needs $cx\mathrm{e}^{x}$ instead. The substitution handles this automatically.

Beyond STEP

The substitution is really a factorisation of the differential operator: writing D for $\mathrm{d}/\mathrm{d}x$, the equation is $(D+3)(D-1)y = 2\mathrm{e}^x$, and $z = (D-1)y$. Factorising operators in this way explains the whole theory of constant-coefficient linear equations, including why repeated or "resonant" roots produce the extra factor of x. Boyce and DiPrima, *Elementary Differential Equations*, develops this systematically.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.